SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name: K.Kabilan		**Reg. No.:** 192521157	
Section / Batch:		**Date:**	24/07/2026

Code of Conduct

I K.Kabilan, Reg. No. 192521157 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: K.Kabilan

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
		GRAND TOTAL:	**100 Marks**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________

SET 1

K.Kabilan | Reg. No. 192521157
Questions 1 - 10

SET 1 — Question 1

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Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Your Fixed Code

```c
#include <stdio.h>

int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while(low <= high) {
        int mid = (low + high) / 2;
        if(arr[mid] == key)
            return mid;
        else if(arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1;
}

int main() {
    int n, key;


    if (scanf("%d", &n) != 1) return 0;

    int arr[n];


    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

```

Test Input

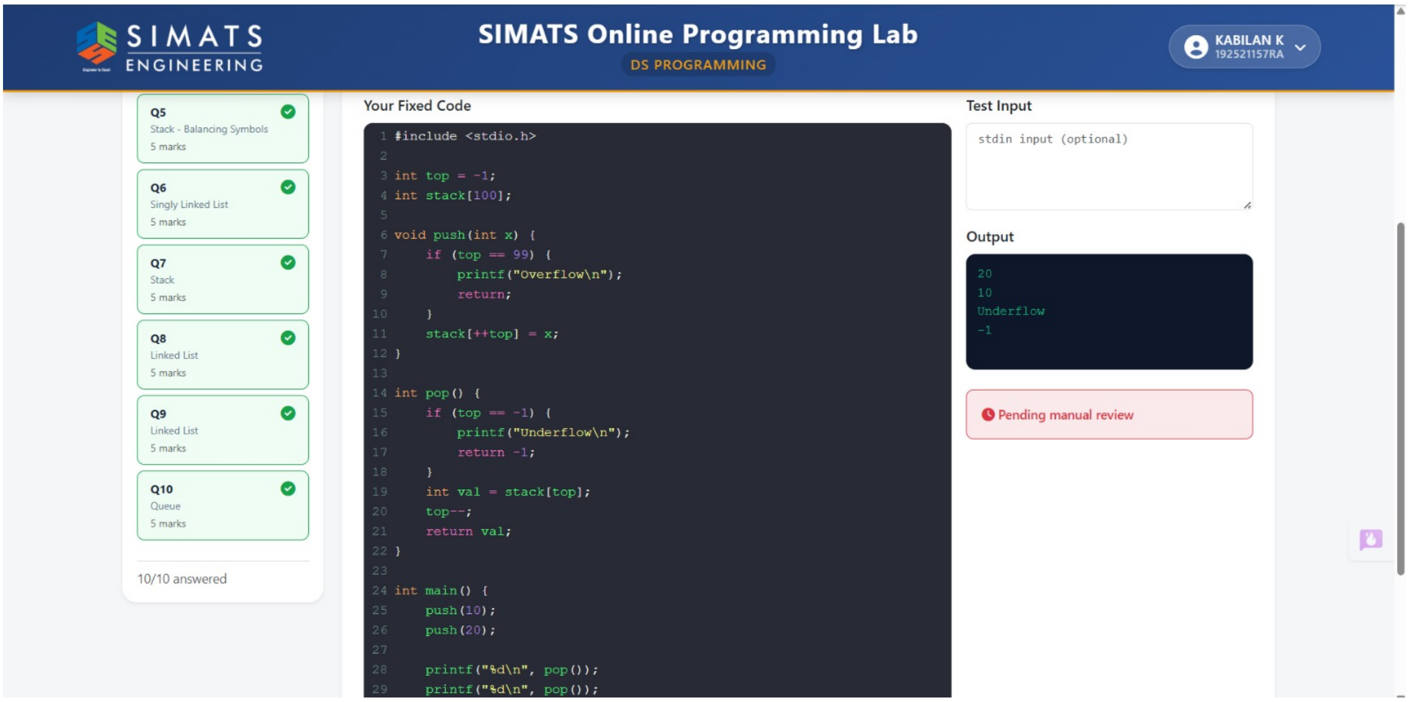
```
7
2 4 6 8 10 12 14
10
```

Output

```
4
```

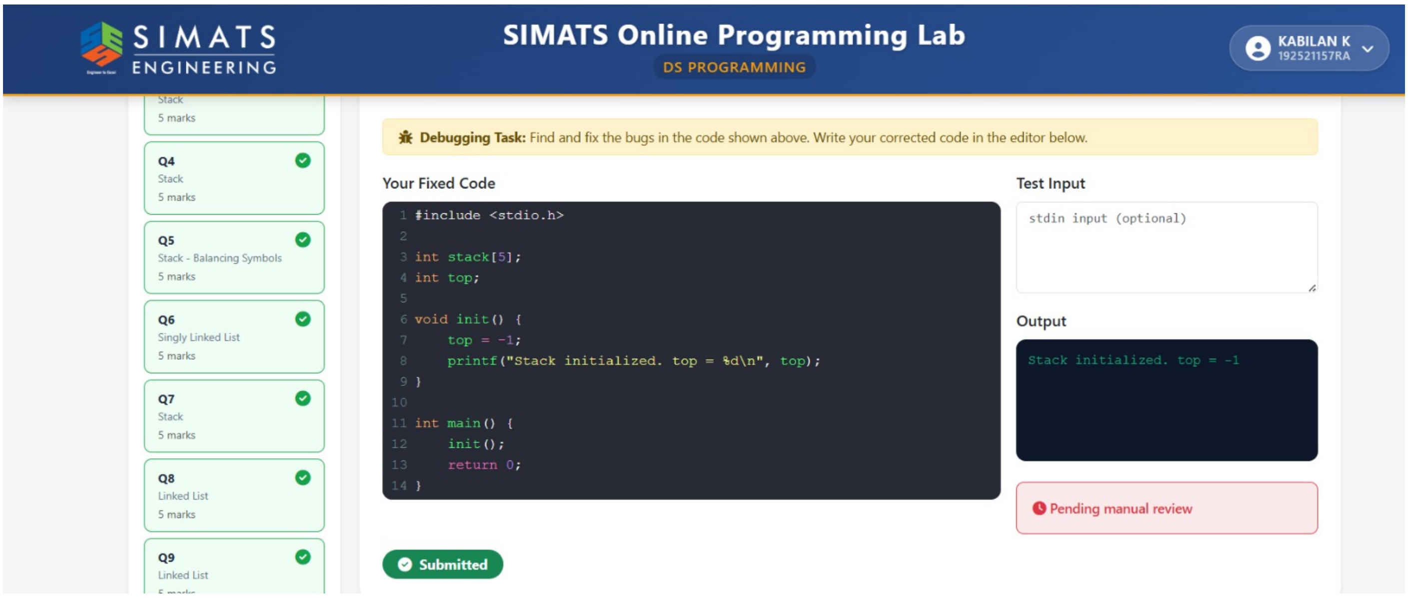
Pending manual review

SET 1 — Question 2

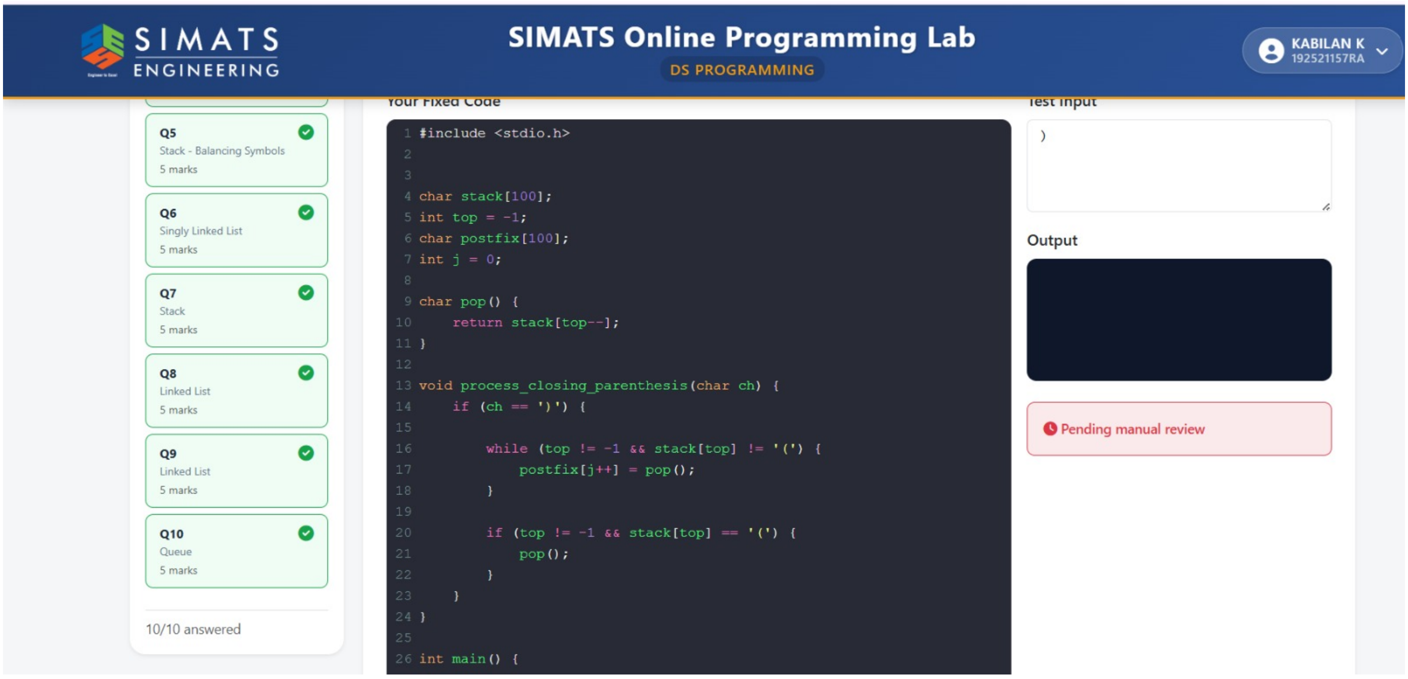



```c
#include <stdio.h>

int top = -1;
int stack[100];

void push(int x) {
    if (top == 99) {
        printf("Overflow\n");
        return;
    }
    stack[++top] = x;
}

int pop() {
    if (top == -1) {
        printf("Underflow\n");
        return -1;
    }
    int val = stack[top];
    top--;
    return val;
}

int main() {
    push(10);
    push(20);

    printf("%d\n", pop());
    printf("%d\n", pop());
```

SET 1 — Question 3

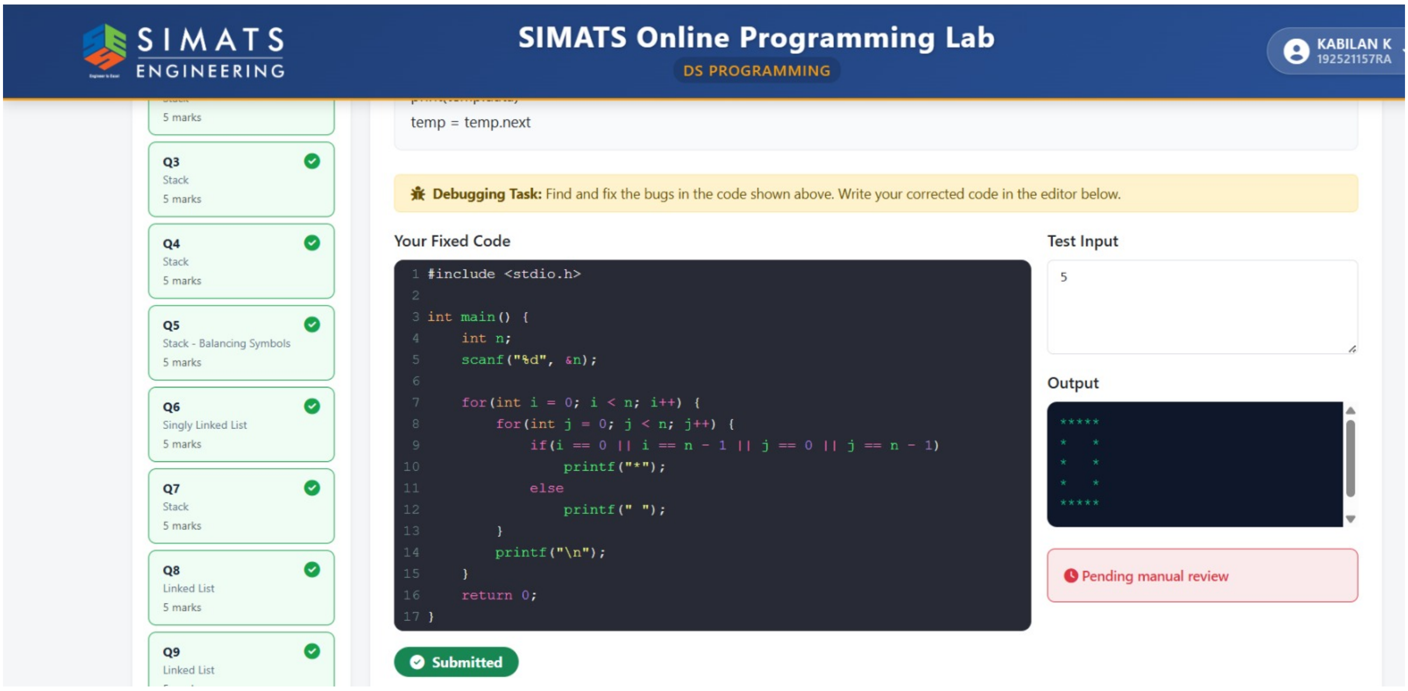


SET 1 — Question 4



SET 1 — Question 5

SET 1 — Question 6



SET 1 — Question 7

SET 1 — Question 8

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Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

typedef struct Node {
    int data;
    struct Node *next;
} Node;

Node* createNode(int x) {
    Node* n = (Node*)malloc(sizeof(Node));
    n->data = x;
    n->next = NULL;
    return n;
}

Node* reverseList(Node* head) {
    Node *prev = NULL, *curr = head, *next = NULL;

    while (curr != NULL) {
        next = curr->next;
        curr->next = prev;
        prev = curr;
        curr = next;
    }
    return prev;
}

void printList(Node* head) {
    Node* temp = head;
```

Test Input

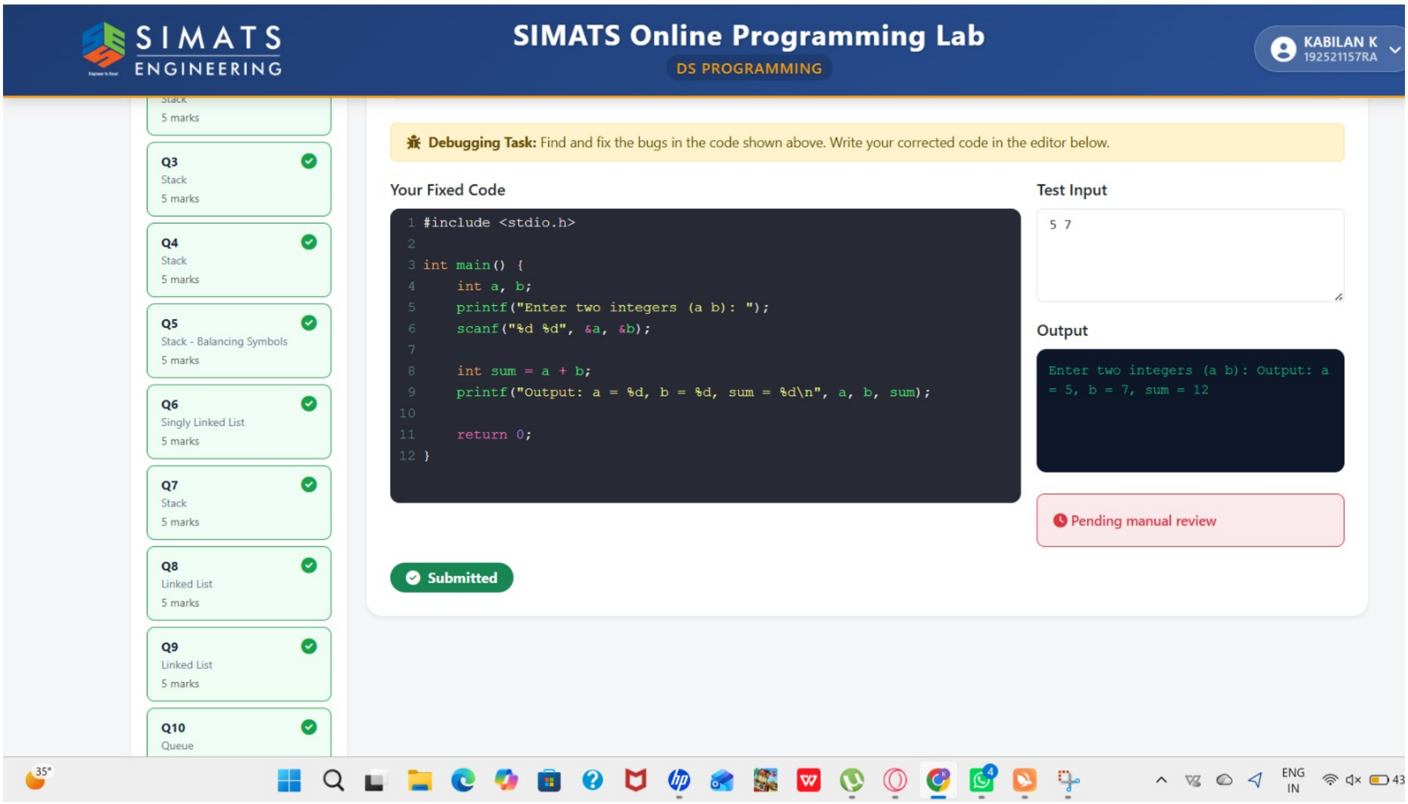
```
5
10 20 30 40 50
```

Output

```
50 40 30 20 10
```

Pending manual review

SET 1 — Question 9



SET 1 — Question 10

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DS PROGRAMMING

KABILAN K
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Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Your Fixed Code

```c
#include <stdio.h>
#define MAX 5

int queue[MAX];
int front = -1;
int rear = -1;

void enqueue(int value) {
    if (rear == MAX - 1) {
        printf("Overflow\n");
    } else {
        if (front == -1) front = 0;
        rear++;
        queue[rear] = value;
    }
}

int dequeue() {
    if (front == -1 || front > rear) {
        printf("Underflow\n");
        return -1;
    } else {
        int item = queue[front];
        front++;

        if (front > rear) {
            front = rear = -1;
        }
        return item;
```

Test Input

```
3
1 10
1 20
1 30
```

Output

```
Dequeued: 10
Dequeued: 20
Dequeued: 30
Dequeued: Underflow
```

Pending manual review

SET 2

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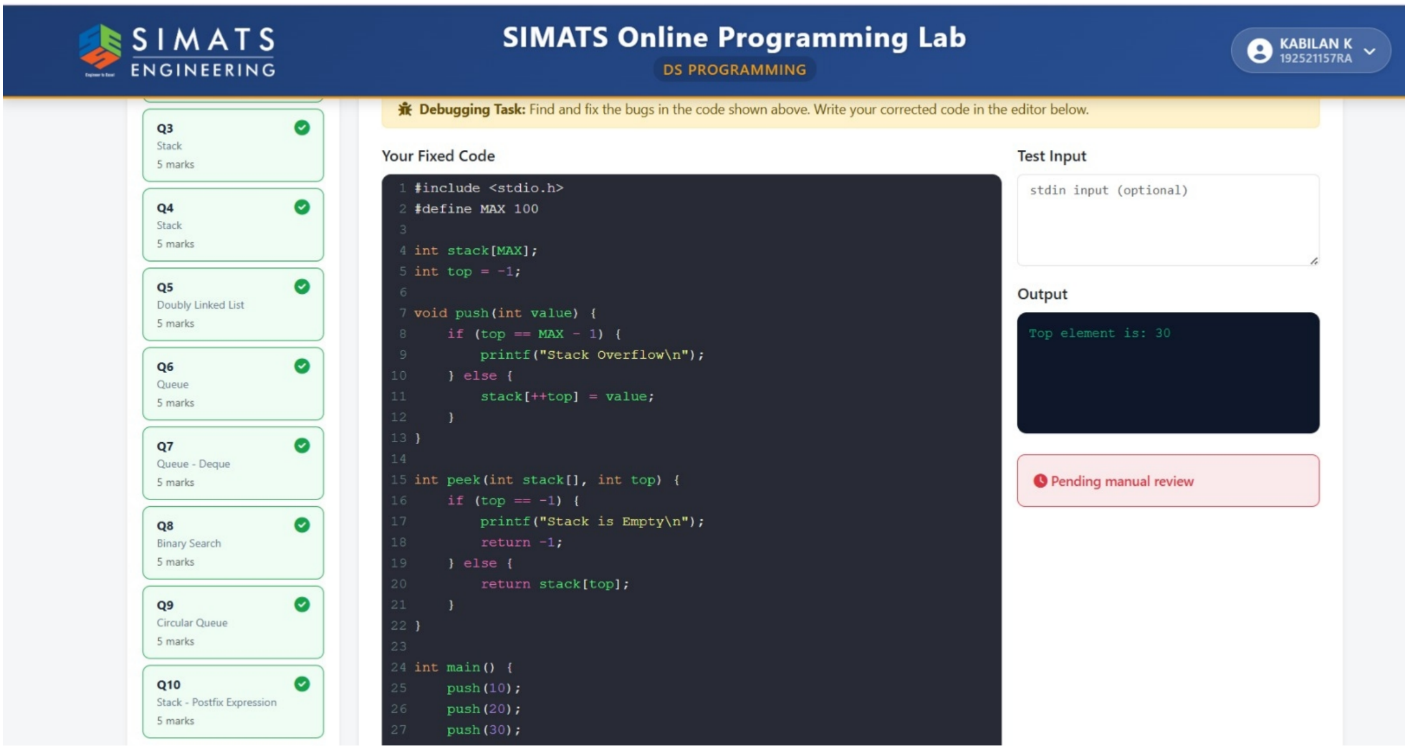
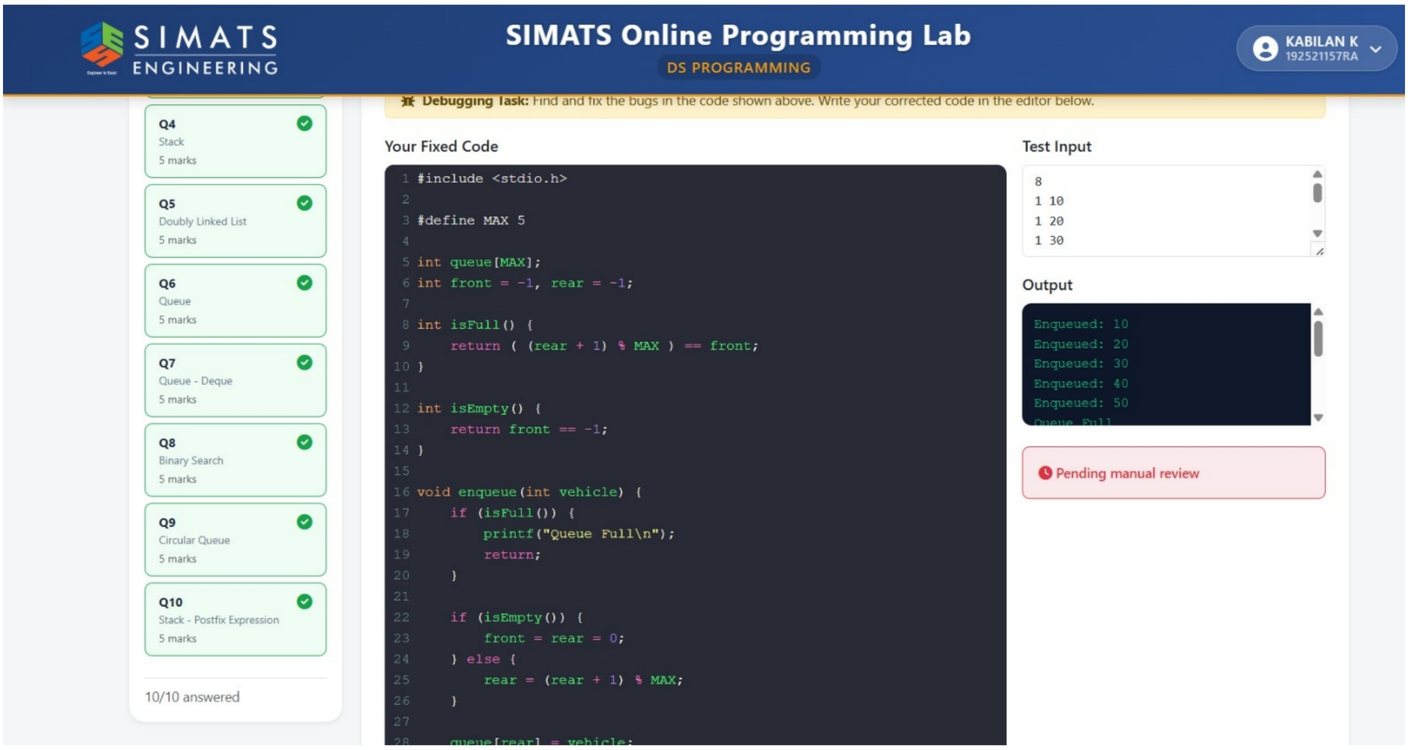
Questions 1 - 10

SET 2 — Question 1



SET 2 — Question 2



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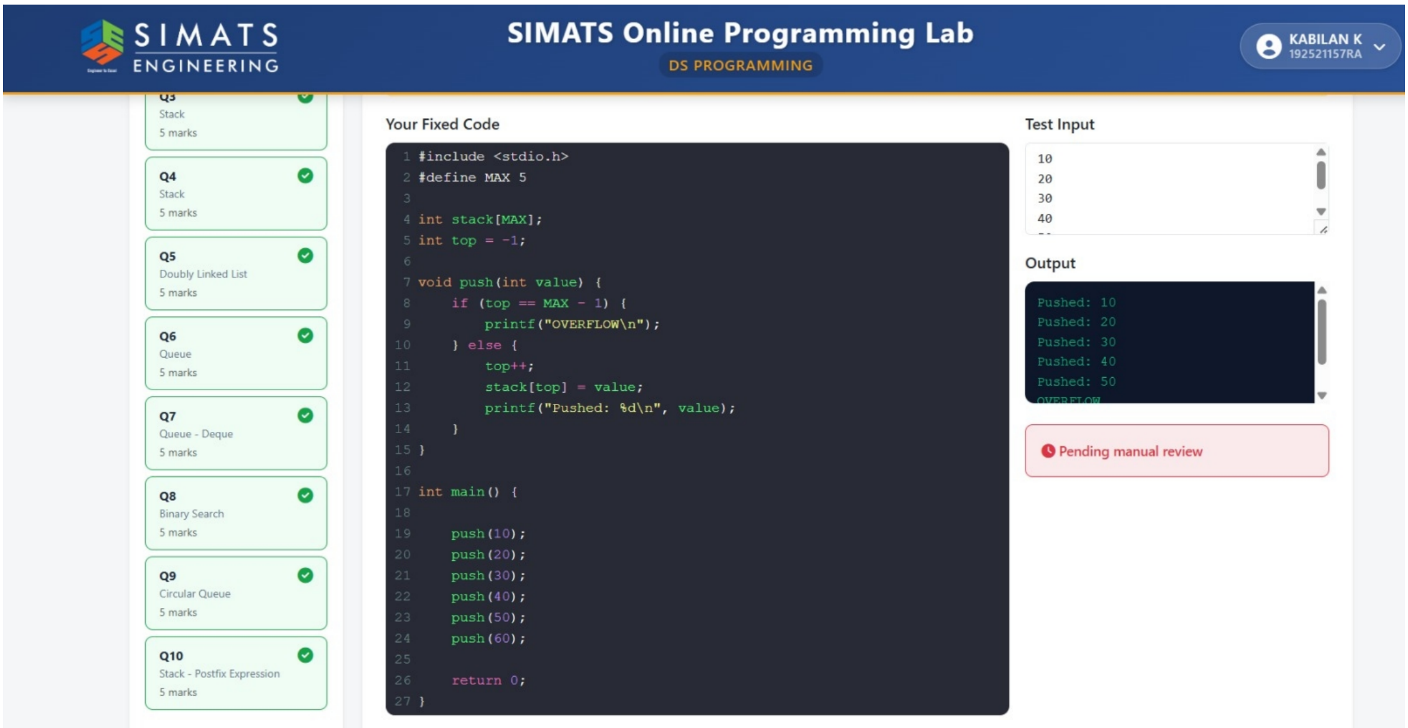
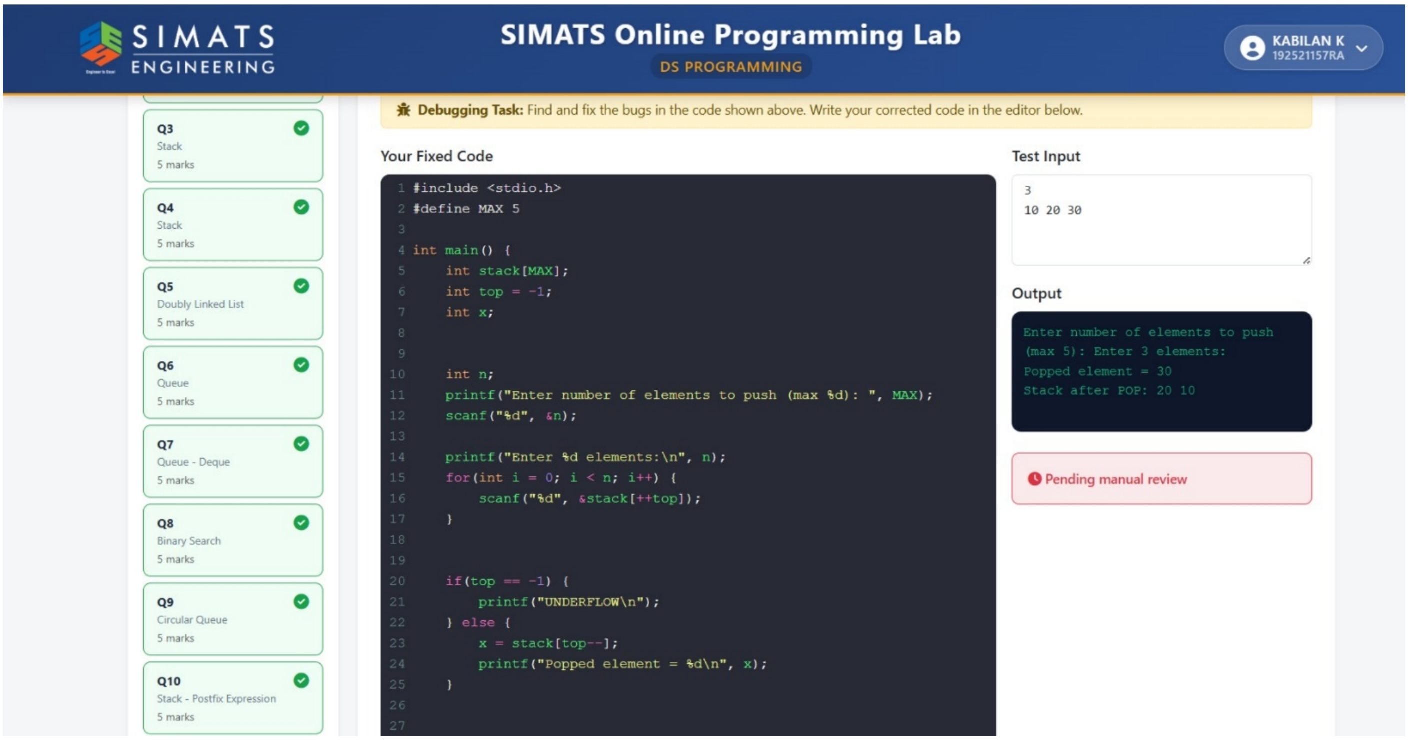
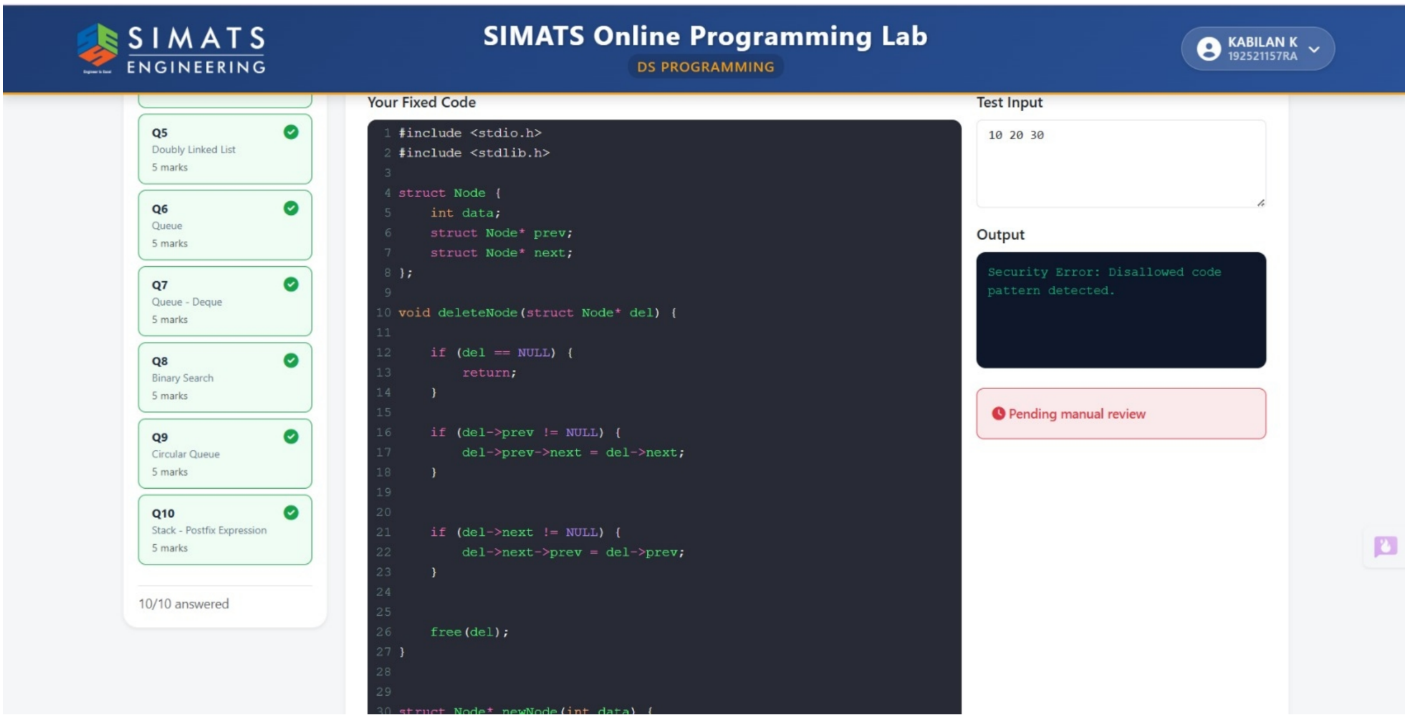


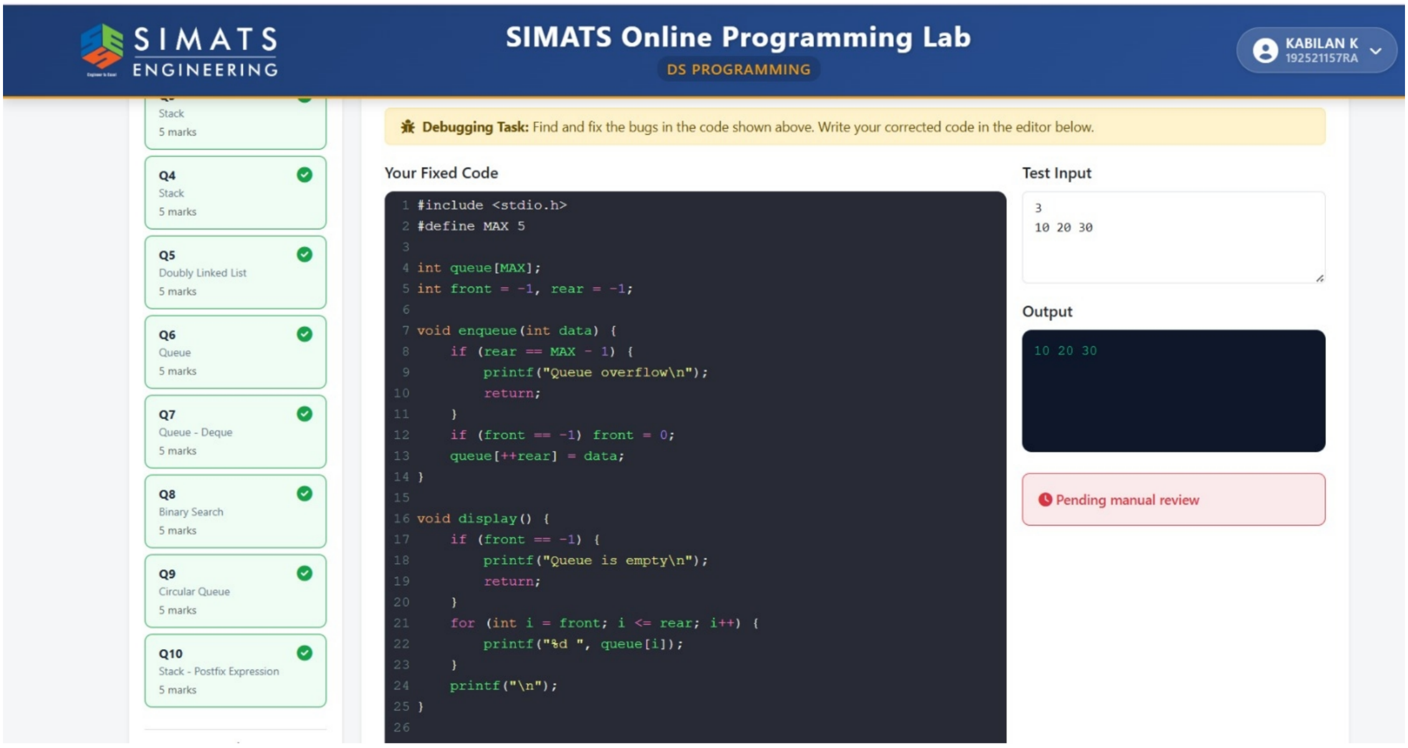
SET 2 — Question 4



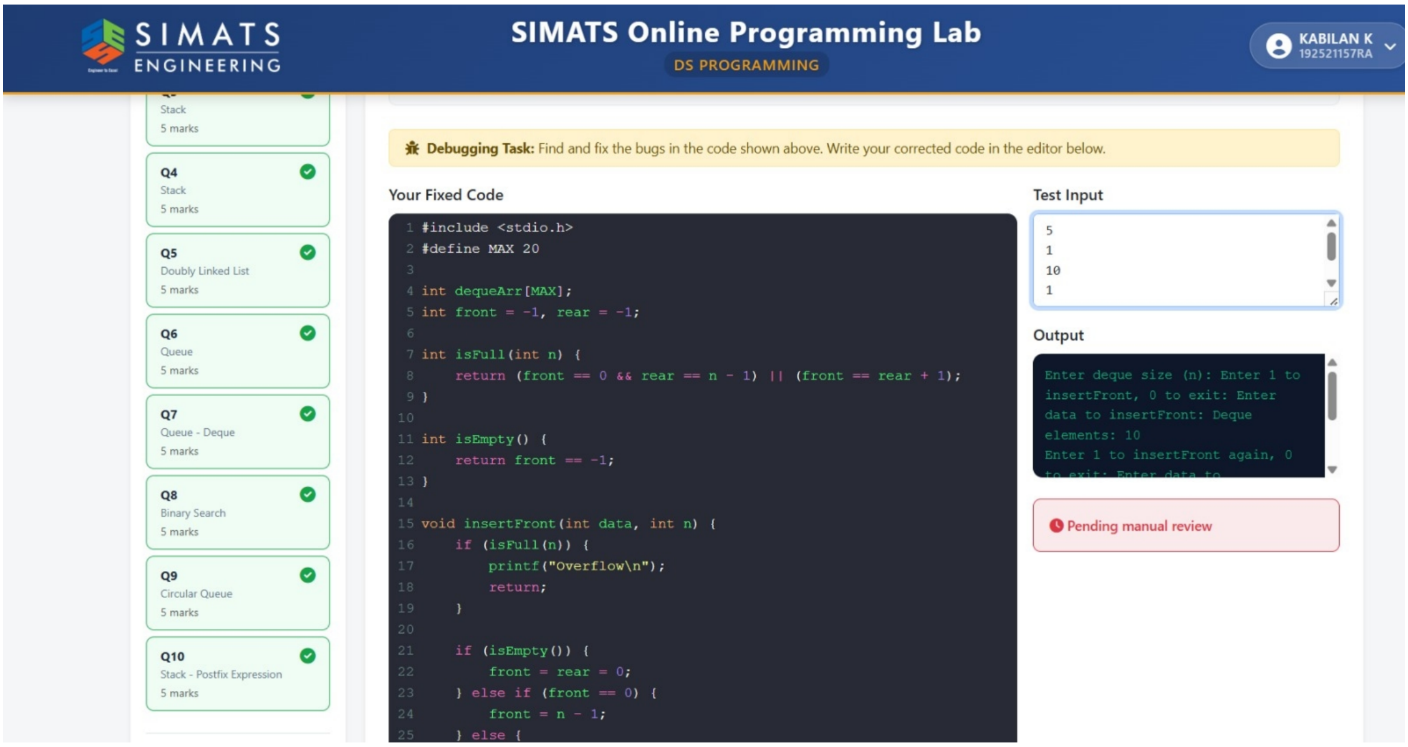
SET 2 — Question 5



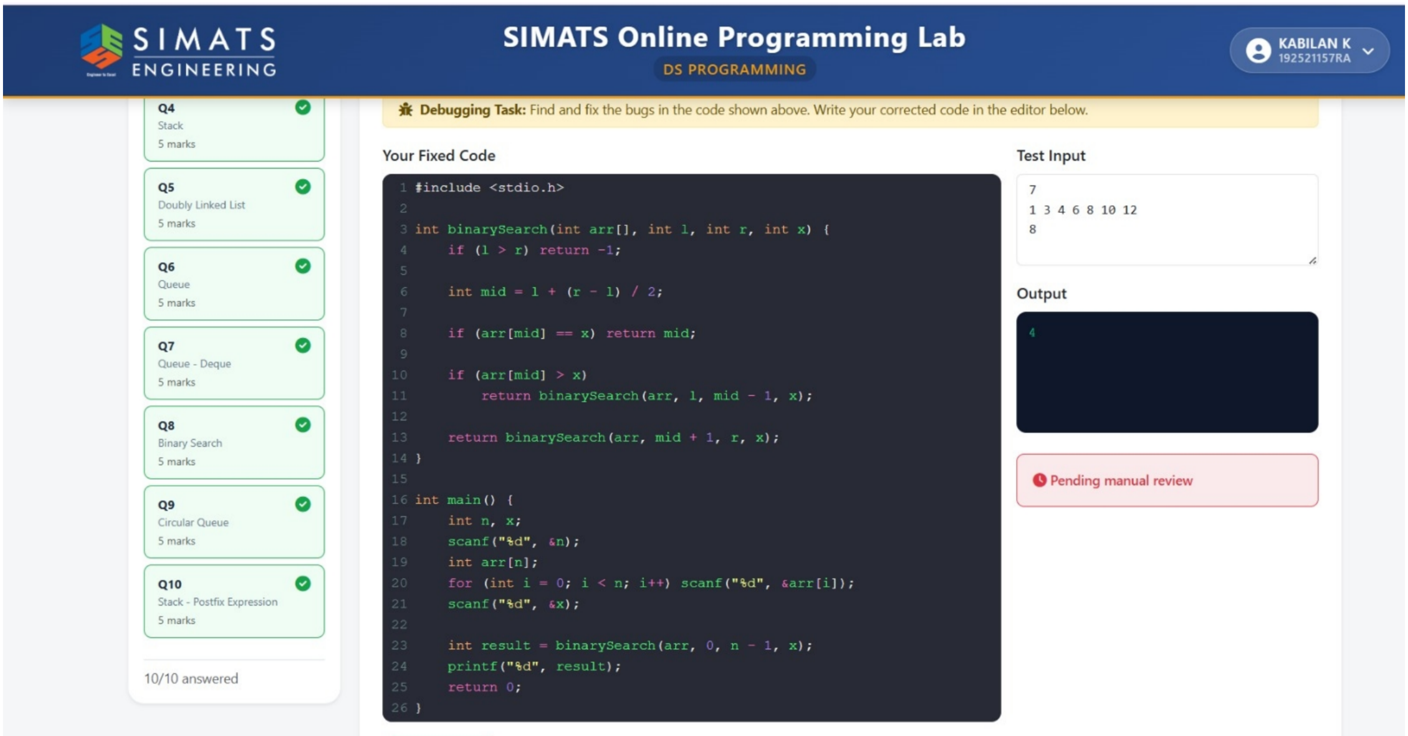
SET 2 — Question 6



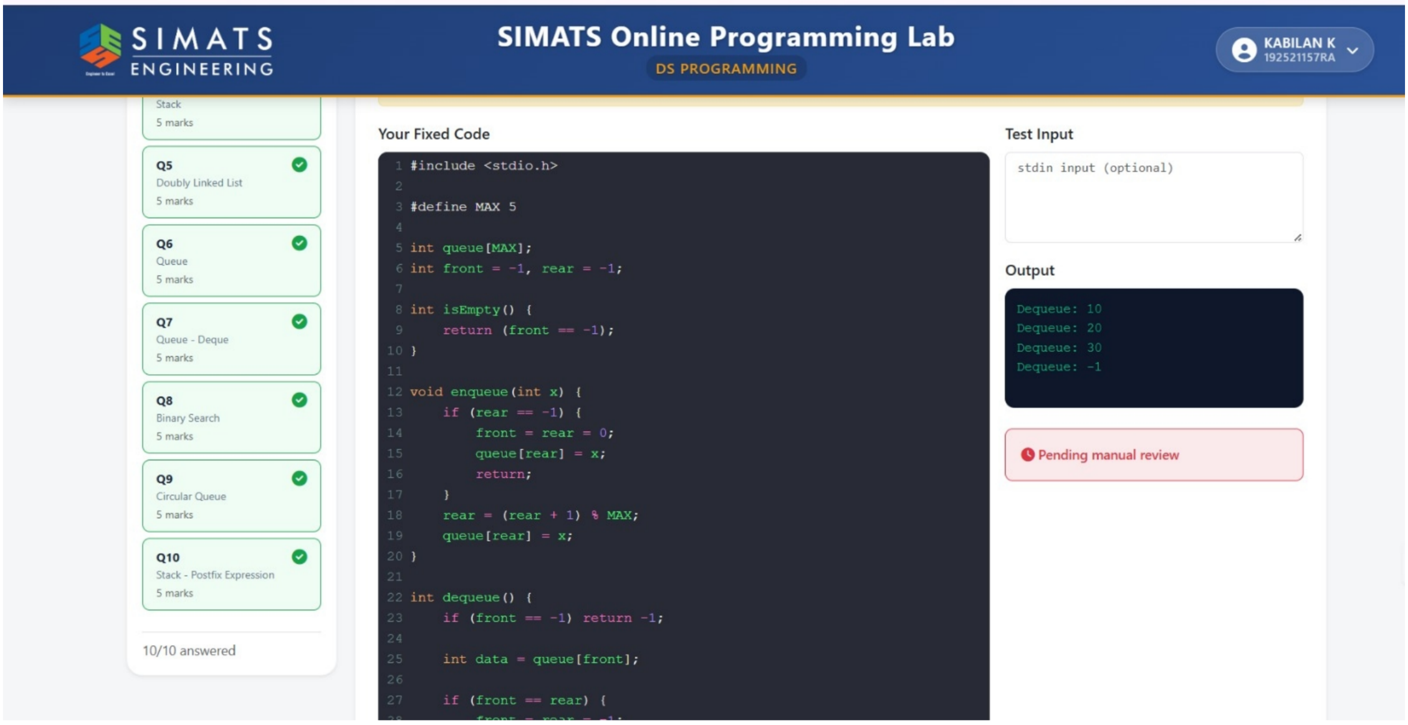
SET 2 — Question 7



SET 2 — Question 8



SET 2 — Question 9



SET 2 — Question 10